

Affirmative Answer for the BDP Status of the Operator S in arXiv:2209.01981

Run fa_banach_001, lane 13

2026-06-16

Status

Candidate full solution, likely valid. This packet answers the local question asked after Example 3.1.1 of Emelyanov–Gorokhova [1]: whether the operator $S : L^1[0, 1] \rightarrow \ell^\infty$ defined there is a [BDP]-operator.

Source Question

The source paper defines

$$S(f) = \left(\int_0^1 f(t) r_k^+(t) dt \right)_{k=1}^\infty, \quad f \in L^1[0, 1],$$

where r_k are the Rademacher functions. After Example 3.1.1, the authors write:

“We do not know whether or not the operator S in Example 3.1.1 is a [BDP]-operator.”

The relevant source crop from page 15 is shown below.

in $[0, 1] \subseteq L^1[0, 1]$, which is an a-limited subset of $L^1[0, 1]$, e.g. by Proposition 3.2.1. The sequence $(Sr_n) = (\frac{1}{2}e_n)$, where e_n are the n -th unite vectors in ℓ^∞ , has no norm convergent subsequences, and hence $S \notin [\text{GPP}](L^1[0, 1], \ell^\infty)$.

We do not know whether or not the operator S in Example 3.1.1 is a [BDP]-operator.

Definitions Used

Following Definition 1.2.2 of [1], an operator $T : X \rightarrow Y$ is a [GPP]-operator if T carries limited subsets of X to relatively compact subsets of Y . It is a [BDP]-operator if T carries limited subsets of X to relatively weakly compact subsets of Y .

The source paper also recalls the classical fact of Bourgain–Diestel [2] that all separable Banach spaces have the Gelfand–Phillips property: every limited subset of such a space is relatively compact.

Proof Intuition

The question is resolved before one uses any special structure of S . The domain $L^1[0, 1]$ is separable. Therefore its limited subsets are already relatively compact in $L^1[0, 1]$. A bounded linear operator sends relatively compact sets to relatively compact sets, and relative compactness is stronger than relative weak compactness. Hence every bounded operator out of $L^1[0, 1]$, including the displayed operator S , is automatically $[GPP]$ and therefore $[BDP]$.

Theorem 1. *The operator*

$$S : L^1[0, 1] \rightarrow \ell^\infty, \quad S(f) = \left(\int_0^1 f(t) r_k^+(t) dt \right)_{k=1}^\infty,$$

from Example 3.1.1 of [1] is a $[BDP]$ -operator. More strongly, it is a $[GPP]$ -operator.

Proof. The space $L^1[0, 1]$ is separable. By the Bourgain–Diestel theorem recalled in [1], every separable Banach space has the Gelfand–Phillips property. Thus every limited subset $A \subseteq L^1[0, 1]$ is relatively compact in $L^1[0, 1]$.

Let $T : L^1[0, 1] \rightarrow Y$ be any bounded linear operator into a Banach space Y , and let $A \subseteq L^1[0, 1]$ be limited. Since A is relatively compact and T is continuous, $T(A)$ is relatively compact in Y . Therefore T is a $[GPP]$ -operator.

Every relatively compact subset of a Banach space is relatively weakly compact, so every $[GPP]$ -operator is a $[BDP]$ -operator. Applying this to $T = S$ gives the desired conclusion. $\square$

Verification and Scope

The argument proves a slightly stronger statement: every bounded linear operator from $L^1[0, 1]$ to any Banach space is $[GPP]$, hence $[BDP]$. No positivity, domination, or Rademacher-specific computation is required.

This also explains the apparent tension in Example 3.1.1 of the source paper. The Rademacher sequence used there is described as an almost limited subset of $L^1[0, 1]$. Such a sequence can obstruct the strong operator class $[s-GPP]$, which tests almost limited subsets, but it is not by itself an obstruction to $[GPP]$, whose definition tests limited subsets. Indeed, since $L^1[0, 1]$ is separable and has the Gelfand–Phillips property, any limited subset of $L^1[0, 1]$ must be relatively compact.

Novelty Check

Before promotion, the run indexes were searched for arXiv id 2209.01981 and the phrases “Operators Affiliated”, “Banach lattice properties”, and “BDP”. Web searches were also made for the exact and near-exact phrases

"We do not know whether or not the operator S" "BDP",

"Operators Affiliated to Banach Lattice Properties" "BDP" "S",

and

"[BDP]-operator" r_k .

No separate later paper answering this exact question was found. The result is therefore recorded as a direct solution extracted from the source definitions and the standard Bourgain–Diestel fact recalled in the source paper.

Human Review Recommendation

Reviewers should check only that the source definitions of $[GPP]$ and $[BDP]$ are being used exactly as printed, namely for limited subsets of the domain. Under those definitions, the proof is immediate. If the authors intended the strong almost-limited class instead, then the source question should be relabelled as a question about $[s\text{-}BDP]$, not $[BDP]$.

References

- [1] E. Emelyanov and S. Gorokhova, *Operators Affiliated to Banach Lattice Properties and the Enveloping Norms*, arXiv:2209.01981, 2022. <https://arxiv.org/abs/2209.01981>
- [2] J. Bourgain and J. Diestel, *Limited operators and strict cosingularity*, Math. Nachr. 119 (1984), 55–58.